

Utility Responsiveness Scoring

Systematizing Utility Knowledge for Datacenter Siting



Power



Scoring



Mapping



AI Agent

SemiAnalysis × Fluidstack Hackathon

8 Dimensions

Comprehensive utility scoring

Conversational AI

Natural language interface

ArcGIS Integration

Map-based visualization

Power to Prefill, Dirt to Decode

The Problem

Datacenter siting isn't constrained by grid capacity information — hyperscalers have access to OASIS, interconnection queues, and internal engineering teams.

The Real Bottleneck: Utility Behavior

Will they negotiate creatively on tariffs? Follow standard interconnection processes? Have internal champions or institutional resistance to large loads?

□ Current State

- Knowledge exists only as **tribal memory**
- Experienced site selectors hoard insights
- No systematic scoring or comparison
- Long, expensive discovery processes

□ URS Solution

- **Numeric scores** per utility
- Structured rationale with evidence
- Conversational pi agent interface
- ArcGIS feature layer visualization

The Solution: URS

Utility Responsiveness Scoring systematizes utility knowledge into actionable intelligence



8-Dimension Scoring

Comprehensive evaluation framework



LLM Enrichment

Extract structured signals from docs



Conversational Agent

Natural language queries via pi



ArcGIS Integration

Map visualization layer

Contextual Scoring

Adjust weights based on your MW requirement, timeline, and sustainability needs

Evidence-Based

Every score backed by source documents: tariff filings, IRPs, rate cases, earnings calls

Always Fresh

Staleness tracking ensures scores reflect current utility posture, not outdated data

From tribal knowledge to systematic intelligence — democratizing datacenter site selection

8 Scoring Dimensions

Each dimension scored 0-100 with confidence levels • All evidence-backed with source citations

1. Large-Load Tariff

Flexible tariff structures, economic development riders, custom rate schedules

2. Interconnection Speed

Historical study completion timelines, fast-track processes

3. IRP Alignment

Planning for datacenter/industrial load growth scenarios

4. Regulatory Environment

State PUC attitudes, incentive legislation, commission precedent

5. Leadership Posture

CEO/COO statements, dedicated large-customer teams

6. Grid Headroom

Capacity constraint handling, speculative transmission builds

7. Track Record

History serving 50+ MW customers, organizational muscle memory

8. Clean Energy Posture

Green tariffs, PPA accommodation, 24/7 CFE programs

Composite Score = Weighted Average by Confidence

$$S = \sum(\text{weight}_i \times \text{score}_i \times \text{confidence}_i) / \sum(\text{weight}_i \times \text{confidence}_i)$$

80-100

Highly Responsive

60-79

Moderately Responsive

40-59

Mixed Signals

20-39

Likely Slow

0-19

Unresponsive

Data Pipeline

Six-stage pipeline from raw documents to scored utilities

0. Source Acquisition

Scheduled crawlers pull from EIA, FERC, state PUCs, SEC EDGAR

1. LLM Extraction

Extract structured signals from unstructured documents

2. Data Merge

Merge LLM extractions with structured sources

3. Dimension Scoring

Apply deterministic heuristics to compute 8 scores

4. Composite Aggregation

Weighted average by confidence, assign tier

5. Rationale Generation

LLM generates narrative from scorecard

Key Data Sources

Structured Sources

EIA-861, FERC Form 1 & 715, OASIS, HIFLD GIS

Regulatory Filings

State PUC dockets, IRPs, rate cases, tariff books

Market Intelligence

SEC 10-K, earnings calls, trade press, LegiScan

On-Demand

Agent request triggers scoring with cache check

Batch Refresh

Scheduled jobs update all utilities with new data

Force Refresh

Manual override bypasses cache, re-runs pipeline

Conversational Agent Interface

Natural language interaction via **pi terminal agent** — no REST APIs, just conversation

Why Conversational?

Site selectors think in questions, not API endpoints. URS meets users where they are — in natural language.

□ "How responsive is Duke Energy Carolinas?"

urs_lookup → urs_score → conversational scorecard

□ "Score Duke for 200MW targeting mid-2027"

Extract context → urs_score with parameters

□ "Compare utilities in Virginia"

urs_lookup each → score → present comparison table

□ "I have Duke's IRP at ./docs/duke-irp.pdf"

urs_ingest → summarize extraction results

6 Custom Tools (Pi Extension)

urs_score

Score a utility

urs_history

Score history

urs_lookup

Resolve utility ID

urs_sources

List data sources

urs_ingest

Ingest source doc

urs_arcgis_sync

Sync to map layer

□ Disambiguation

Agent asks clarifying questions for ambiguous queries

□ Context Extraction

Extracts MW, timeline, use case from natural language

□ Guided Exploration

Agent suggests weight adjustments based on priorities

ArcGIS Visualization Layer

Scores rendered as **feature layer** over utility service territory polygons — visual spatial intelligence

□ Territory Mapping

- Service territory polygons from **HIFLD**
- Each polygon = one utility EIA ID
- Graduated color by composite tier (1-5)
- Click for full scorecard popup

□ Attribute Layer

- 28 flattened attributes per utility
- Composite score + 8 dimension scores
- Confidence, tier, data vintage, staleness flags
- Queryable via SQL WHERE clauses

□ Symbology

- Tier 1 (80-100): Dark Green
- Tier 2 (60-79): Light Green
- Tier 3 (40-59): Yellow
- Tier 4 (20-39): Orange
- Tier 5 (0-19): Red

□ Sync Workflow

- Agent calls `urs_arcgis_sync`
- Pushes scores via ArcGIS REST API
- Updates feature layer in real-time
- Map viewers see latest scores instantly

Example Map Queries

```
URS_TIER <= 2 AND STATE = 'VA'
```

Virginia utilities with high responsiveness

```
DIM_CLEAN > 70
```

Utilities strong in clean energy posture

Technical Architecture

Built on **pi coding agent** with TypeScript extensions and Claude 3.5 Sonnet

□ Agent Layer

- **Pi terminal agent** (.pi/extensions/urs/)
- 6 custom tools registered via ExtensionAPI
- URS skill (.pi/skills/urs/SKILL.md)
- Natural language → tool call orchestration

□ Pipeline Layer

- Scheduled crawlers (Python/TypeScript)
- Deterministic scoring heuristics
- Caching with TTL based on staleness
- Batch and on-demand execution modes

□ LLM Layer

- **Claude 3.5 Sonnet** for extraction
- Temperature 0 for structured extraction
- JSON schema validation on outputs
- Temperature 0.3 for rationale generation

□ Integration Layer

- **ArcGIS REST API** for feature service
- applyEdits for real-time sync
- HIFLD service territory polygons
- 28-attribute flattened schema

□ Data Layer

- PostgreSQL for structured scoring DB
- Blob storage for raw source documents
- Version tracking for all extractions
- Staleness metadata per source

□ Security Layer

- API key management for data sources
- Rate limiting on external APIs
- Source provenance for all extractions
- Audit trail for score computations

Use Cases & Impact

Real-world applications across the datacenter development lifecycle

□ Site Screening

Filter 200+ US utilities down to top 20 responsive candidates in minutes, not months

Value: \$500K-2M in consultant fees saved

□ Comparative Analysis

"Compare Duke, Dominion, and AEP for 300MW hyperscale" — instant evidence-based comparison

Value: 6-8 weeks timeline compression

□ Due Diligence

Deep-dive rationale with source citations validates investment committee presentations

Value: Risk reduction in \$500M+ projects

□ Geographic Prioritization

Overlay URS scores with fiber routes and land parcels in ArcGIS for holistic site ranking

Value: Multi-criteria optimization

□ Market Intelligence

Track score history to detect regulatory shifts, leadership changes, new tariff approvals

Value: First-mover advantage

□ Utility Engagement

Enter negotiations armed with evidence: "Your IRP mentions datacenter growth — let's discuss"

Value: Stronger negotiating position

□ Total Addressable Impact

\$50B+

Annual datacenter capex in US (2026)

6-12 months

Typical site selection timeline compression

200+ utilities

US IOUs covered (80%+ of load)

Hackathon Fit: Dirt to Decode

Theme: Power to Prefill, Dirt to Decode, Transformers to Transformers

□ Dirt Layer

URS starts at the dirt: Land permitting begins with utility responsiveness. Can you interconnect? Will they negotiate? URS answers before breaking ground.

□ Power Layer

Power to Prefill: Scoring utilities on tariff flexibility, interconnection speed, and grid headroom — the power delivery infrastructure AI needs.

□ Transformers Layer

Transformers to Transformers: Electrical transformers enable LLM transformers. URS uses Claude to extract utility posture from docs.

Why URS Perfectly Fits This Hackathon

□ Full-Stack AI Infrastructure

Not just the API layer — tackles **land, power, and siting** as first-order problems for AI infrastructure

□ Real Economic Impact

\$50B+ annual datacenter capex — **6-12 month timeline compression** = billions in opportunity cost saved

□ Novel Data Pipeline

LLM-driven extraction from IRPs, rate cases, earnings calls — **unstructured** → **structured intelligence**

□ Innovative UX

Conversational agent via **pi terminal** — not REST APIs, natural language queries with contextual scoring

"From dirt to decode: URS reimagines datacenter site selection by systematizing the most opaque variable — utility behavior — into evidence-based intelligence."

Limitations & Future Work

Honest assessment of current constraints and research directions

❑ Private Negotiation Opacity

Filed tariffs show public offers; negotiated contracts often confidential. May undercount utilities that negotiate aggressively but privately.

Mitigation: Lower confidence when no special contracts visible

❑ Regulatory vs. Utility Confounding

Low scores may reflect state regulatory constraints rather than utility reluctance. Hard to isolate.

Mitigation: Separate regulatory dimension; allow zero-weighting

❑ Small Utility Blindness

Municipal utilities and co-ops file less with FERC/PUCs. Limited public data → low confidence scores.

Mitigation: Coverage warnings; focus on ~200 IOUs (80%+ load)

❑ Geographic Uniformity Assumption

Scores are utility-level, not substation-level. Duke may be responsive in Charlotte, unresponsive in rural areas.

Future: Add lat/lon to intersect with transmission infrastructure

❑ LLM Extraction Errors

Misinterpretation of tariff language or missed IRP sections can propagate to scores.

Mitigation: Schema validation; spot-checks; extraction confidence

❑ Temporal Lag

Utility behavior changes (new CEO, rate case) before scoring system detects. Inherent reactive lag.

Mitigation: Staleness tracking; alerts for high-impact events

❑ Future Research Directions

Substation-level granularity: Score specific interconnection points, not just utilities

Sentiment analysis: Track tone shifts in earnings calls and PUC testimony over time

Predictive models: Forecast score changes based on regulatory/leadership signals

Demo Walkthrough

Sample interaction showing conversational URS in action

user@pi:~\$

"How responsive is Duke Energy Carolinas for a 200MW datacenter targeting mid-2027?"

[Agent calls urs_lookup("Duke Energy Carolinas") → resolves to EIA 6452]

[Agent calls urs_score(utility_id="6452", mw_requirement=200, target_isd="2027-06-01")]

URS Agent:

Duke Energy Carolinas scores **72/100** (Tier 2: Moderately Responsive)

Strengths:

- Strong tariff flexibility (85/100) - Approved LGS and Economic Development Rider
- Excellent track record (88/100) - Serves multiple large loads in Charlotte metro
- High clean energy posture (78/100) - Green Source Advantage program available

Concerns:

- Interconnection speed moderate (62/100) - 14-18 month study timelines historically
- Your 2027-06 timeline is aggressive given typical study pace

Recommendation: Explore phased delivery or interim solutions while upgrades are built.

Confidence: 81% (some data staleness in interconnection dimension)

user@pi:~\$

"Compare that to Dominion Energy Virginia"

[Agent calls urs_lookup("Dominion Energy Virginia") → resolves to EIA 56270]

Utility Responsiveness Scoring

From Tribal Knowledge to Systematic Intelligence



8-Dimension Framework

Evidence-backed scoring with confidence levels



Conversational Agent

Natural language via pi terminal, not REST APIs



ArcGIS Integration

Spatial visualization over service territories

Impact: \$50B+ Annual Datacenter Capex

6-12 month timeline compression • \$500K-2M in consultant fees saved • Risk reduction in \$500M+ projects

Ready to Transform Datacenter Site Selection?

■ Built for SemiAnalysis × Fluidstack

■ Dirt to Decode, Power to Prefill

Making the invisible visible. Systematizing utility responsiveness for the AI infrastructure era.